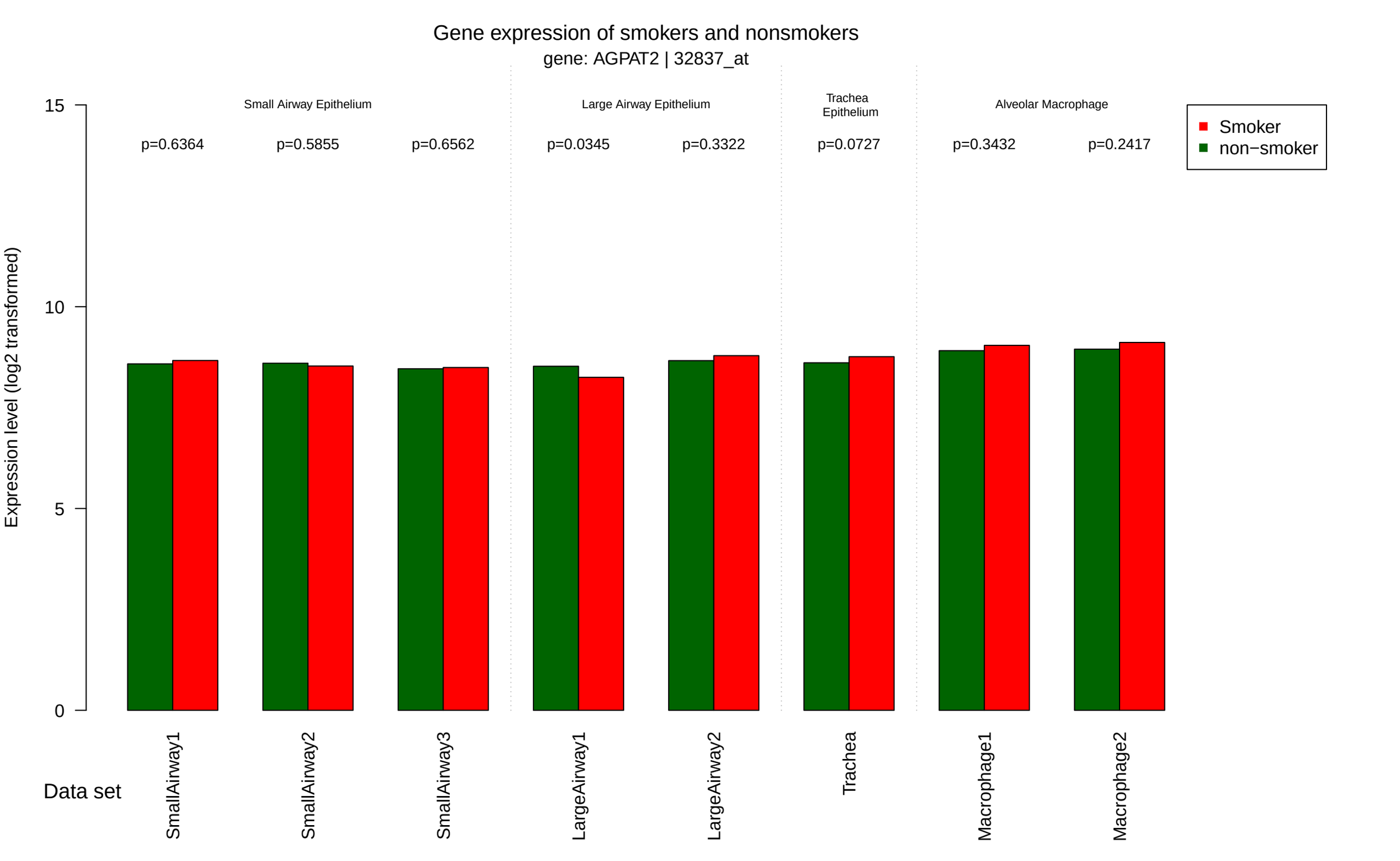
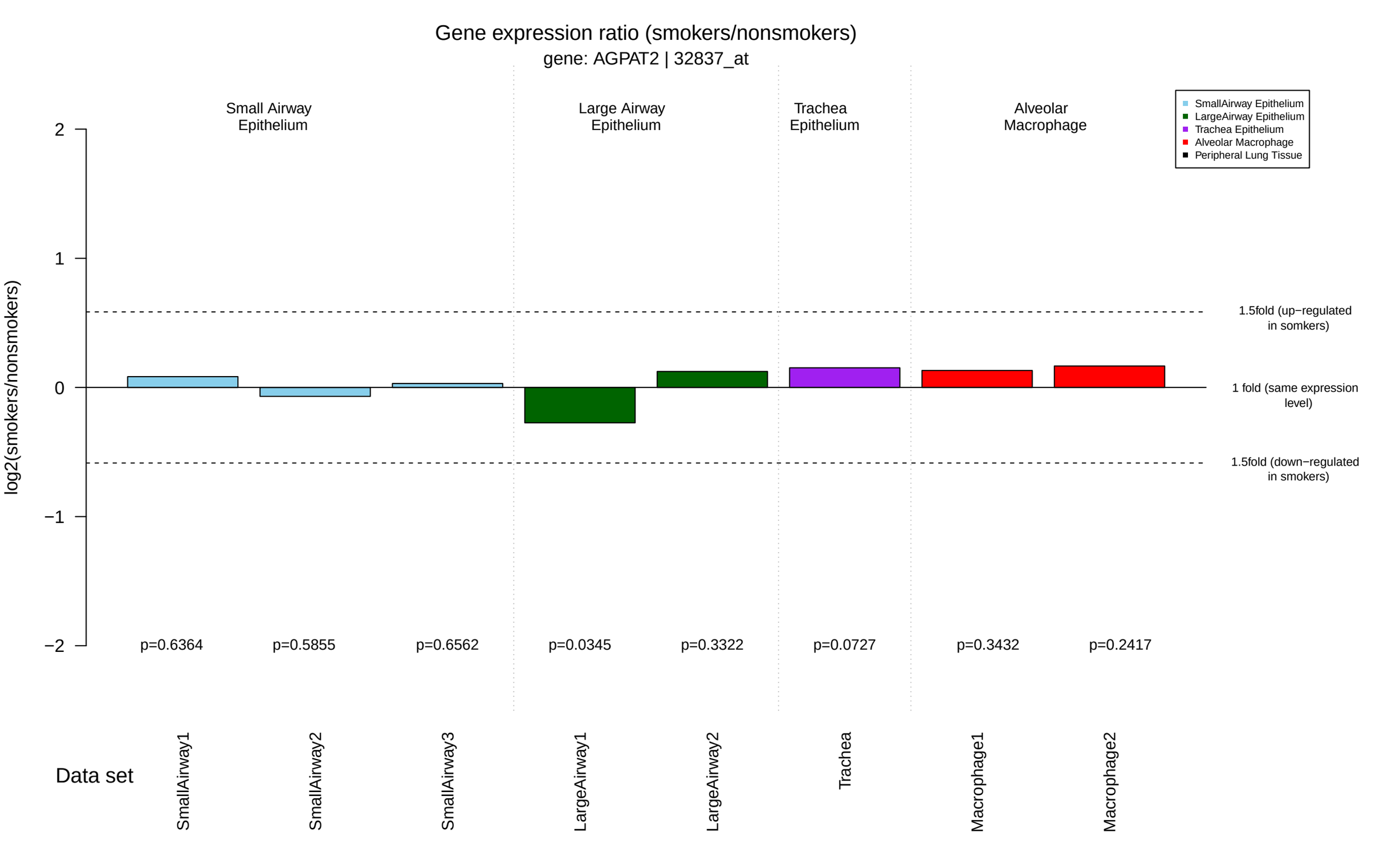
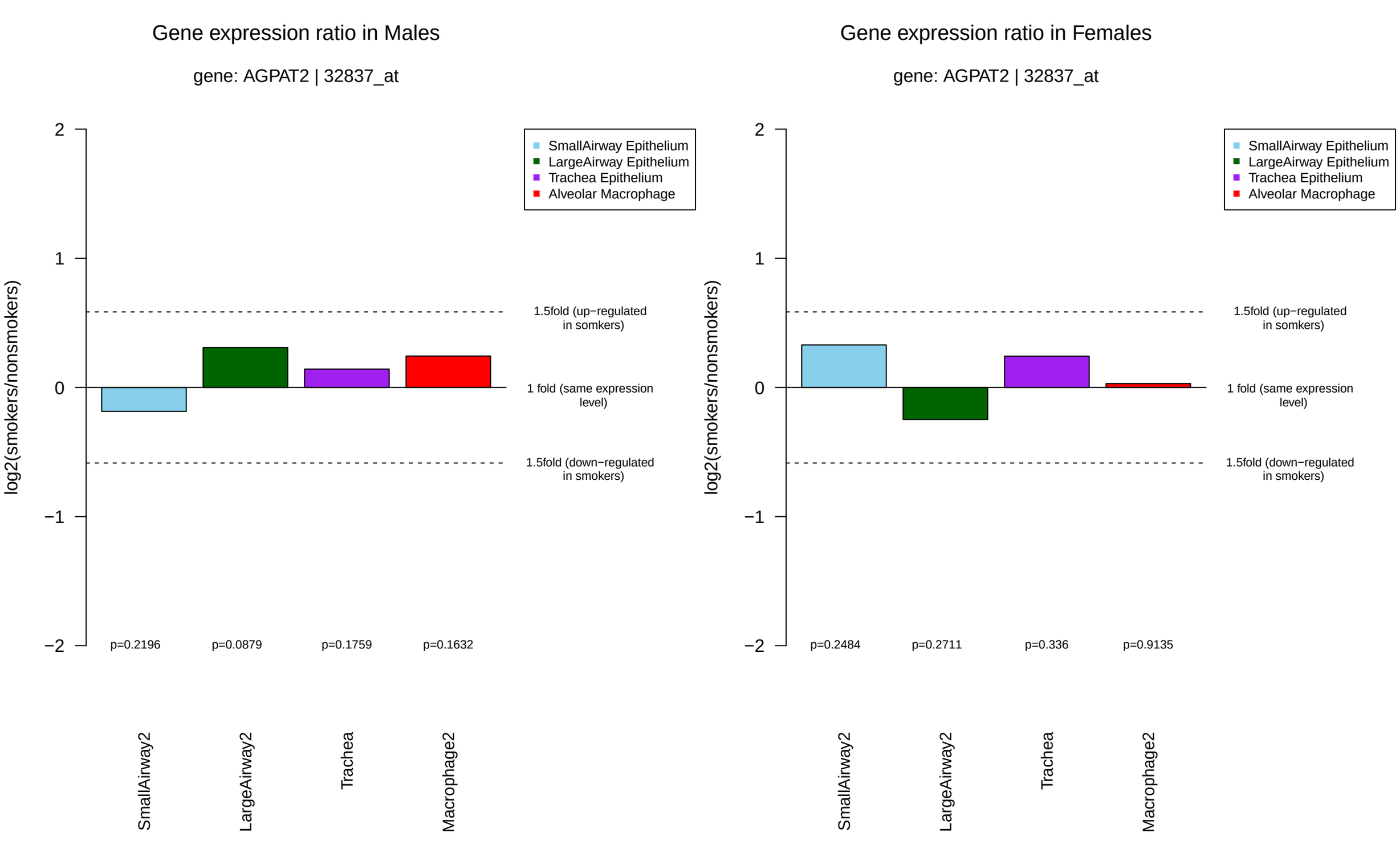
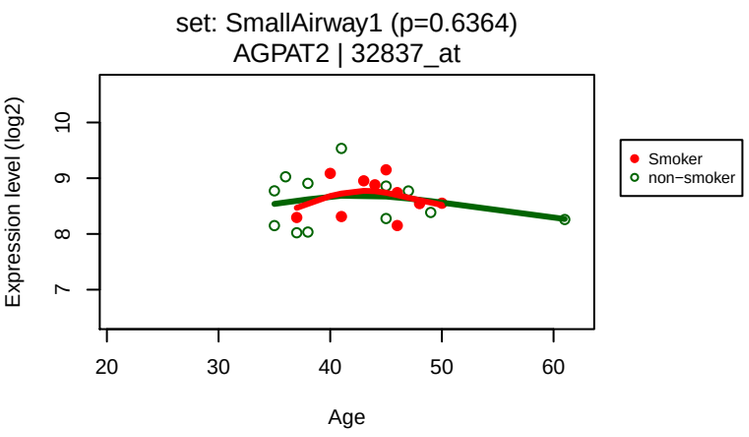
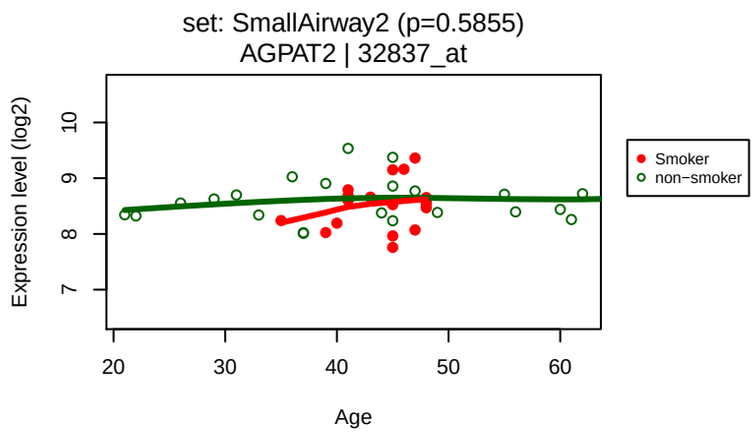
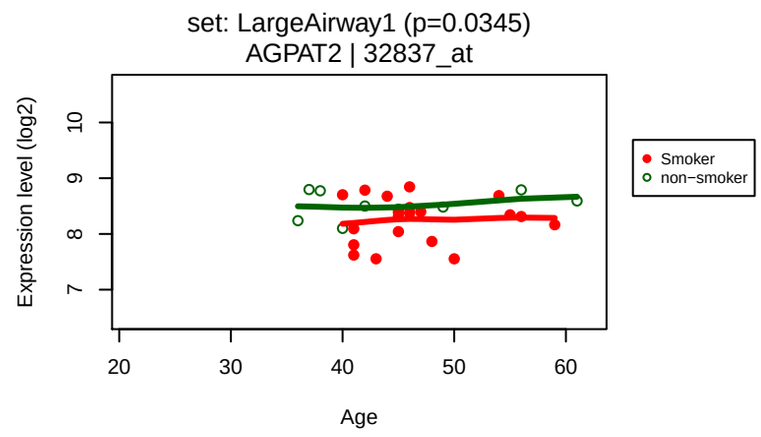

Gene expression of smokers and nonsmokers
gene: AGPAT2 | 32837_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6364
p=0.5855
p=0.6562
p=0.0345
p=0.3322
p=0.0727
p=0.3432
p=0.2417
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

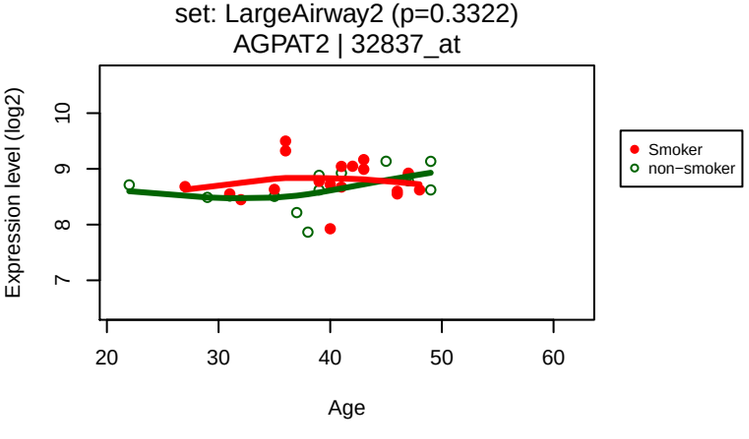

Gene expression ratio (smokers/nonsmokers)
gene: AGPAT2 | 32837_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6364
p=0.5855
p=0.6562
p=0.0345
p=0.3322
p=0.0727
p=0.3432
p=0.2417
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

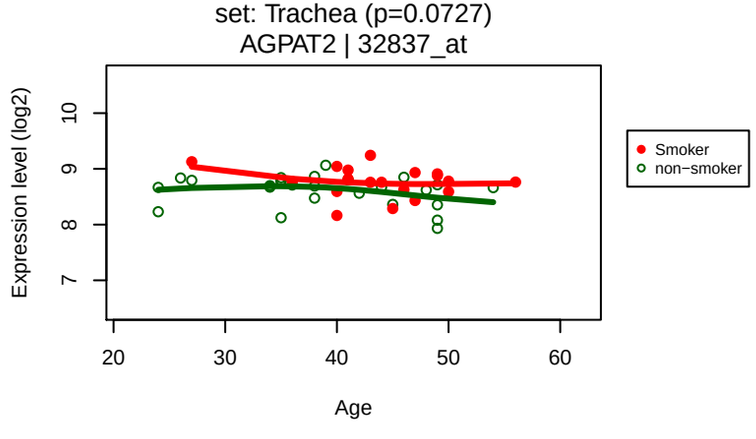

Gene expression ratio in Males
gene: AGPAT2 | 32837_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2196
p=0.0879
p=0.1759
p=0.1632
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AGPAT2 | 32837_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2484
p=0.2711
p=0.336
p=0.9135
SmallAirway2
LargeAirway2
Trachea
Macrophage2

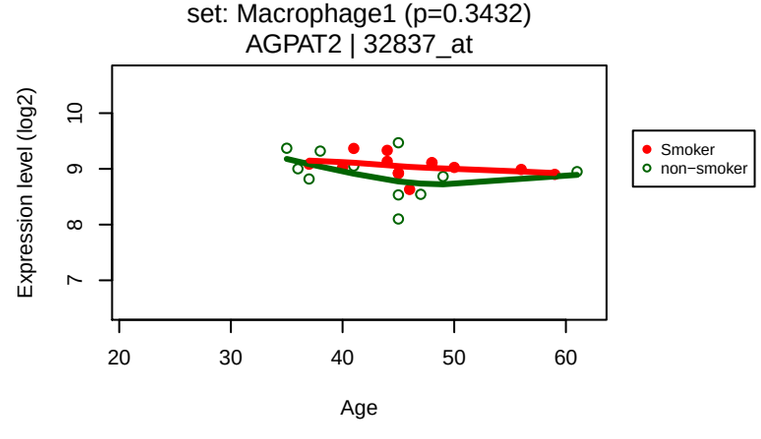

set: SmallAirway1 (p=0.6364)
AGPAT2 | 32837_at
Expression level (log2)
Age
Smoker
non−smoker

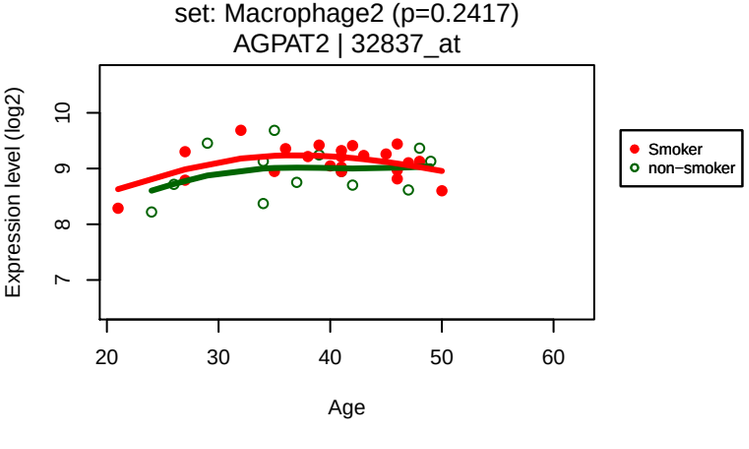

set: SmallAirway2 (p=0.5855)
AGPAT2 | 32837_at
Expression level (log2)
Age
Smoker
non−smoker

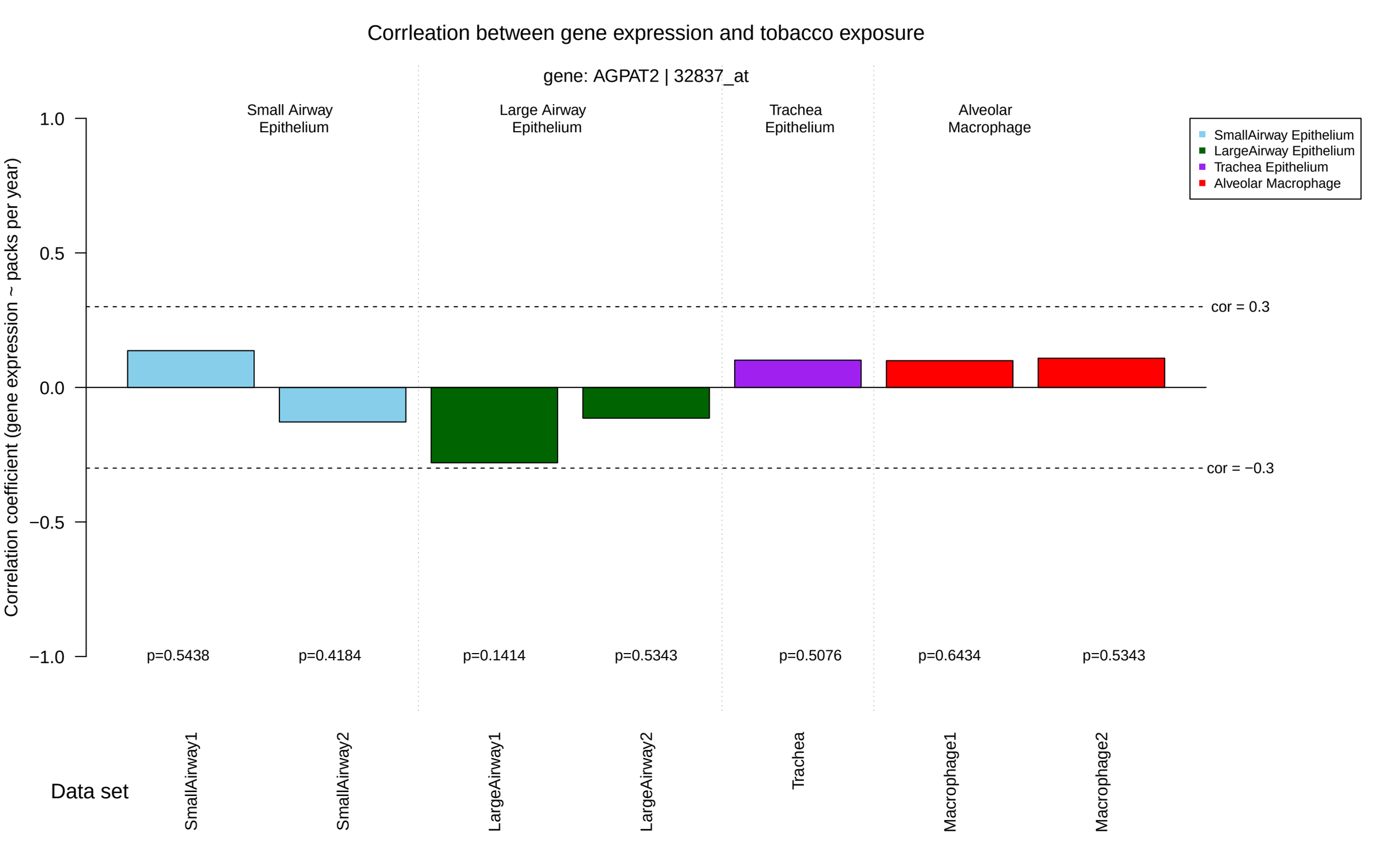

set: LargeAirway1 (p=0.0345)
AGPAT2 | 32837_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3322)
AGPAT2 | 32837_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0727)
AGPAT2 | 32837_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3432)
AGPAT2 | 32837_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2417)
AGPAT2 | 32837_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AGPAT2 | 32837_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5438
p=0.4184
p=0.1414
p=0.5343
p=0.5076
p=0.6434
p=0.5343
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2